

Answers:

1. True/False: To add 2 matrices together, they must be the same size.
 - a. True
2. True/False: When you scale a matrix by a scalar value, the result is scaling the dimensions of the matrix.
 - a. False, you simply multiply each element but the dimensions remain the same.
3. True/False: To multiply 2 matrices together, they must be the same size.
 - a. False, if you multiply 2 matrices together, A and B, the # of columns of A must match the # of rows of B.
4. True/False: In matrix-vector multiplication, the vector can be thought of as an nx1 matrix.
 - a. True
5. True/False: Multiplying a matrix by the identity matrix, yields the identity matrix.
 - a. False, it yields the original matrix
6. True/False: The number of rows in the resulting matrix from a multiplication is determined by the number of rows in the first matrix.
 - a. True
7. Given A = 2x2 matrix, B = 2x2 matrix, is addition/subtraction/multiplication possible? Which? And why?
 - a. The matrices are the same size, so addition/subtraction are possible. The # of columns of A matches the # of rows in B, so multiplication is possible.
8. Given A = 3x2 matrix, B = 2x3 matrix, is addition/subtraction/multiplication possible? Which? And why?
 - a. The matrices are NOT the same size, so neither addition nor subtraction are possible. The # of columns of A matches the # of rows in B, so multiplication is possible.
9. Given A = 2x3 matrix, B = 2x3 matrix, is addition/subtraction/multiplication possible? Which? And why?
 - a. The matrices are the same size, so addition/subtraction are possible. The # of columns of A does NOT match the # of rows in B, so multiplication is not possible.
10. Given A = 5x3 matrix, B = 3x2 matrix, AB will be what dimension?
 - a. 5x2
11. Compute the following:

$$\begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix} + \begin{pmatrix} 5 & 7 \\ 6 & 8 \end{pmatrix}$$

Answer: $\begin{pmatrix} 6 & 10 \\ 8 & 12 \end{pmatrix}$

12. Compute the following:

$$\begin{pmatrix} 10 & 8 \\ 9 & 7 \end{pmatrix} - \begin{pmatrix} 5 & 7 \\ 6 & 8 \end{pmatrix}$$

Answer: $\begin{pmatrix} 5 & 1 \\ 3 & -1 \end{pmatrix}$

13. Compute the following:

$$\begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix} + \begin{pmatrix} 5 & 7 & 9 \\ 6 & 8 & 10 \end{pmatrix}$$

Answer: These matrices are different dimensions, and thus cannot be added or subtracted.

14. Compute the following:

$$6 \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix}$$

Answer: $\begin{pmatrix} 6 & 18 \\ 12 & 24 \end{pmatrix}$

15. Compute the following:

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \times \begin{pmatrix} 5 & 7 \\ 6 & 8 \end{pmatrix}$$

Answer: $\begin{pmatrix} 5 & 7 \\ 6 & 8 \end{pmatrix}$

16. Compute the following:

$$\begin{pmatrix} 1 & 1 \\ 2 & 1 \end{pmatrix} \times \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix}$$

Answer: $\begin{pmatrix} 3 & 7 \\ 4 & 10 \end{pmatrix}$

17. Is matrix addition commutative?

a. Yes

18. Associative?

a. Yes